ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Resources and Development (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** stock	Water is an example (H2 and O2)
2	**(b)** Rocks	Non-living
3	**(b)** community-owned resources	Community
4	**(a)** rivers	Indus, Ganga, Brahmaputra
5	**(d)** resource	Needs technology and economy
6	**(b)** renewable resources	Solar, wind
7	**(a)** Mahatma Gandhi	Quote
8	**(c)** shelter belts	Dry regions
9	**(a)** arid soils	Sandy and saline
10	**(b)** khadar	Lowland, fertile

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	On the basis of exhaustibility, resources are renewable (solar and wind energy, water, forests) (1); and non-renewable (minerals and fossil fuels, which take millions of years to form) (1).	1+1
12	A resource is anything in our environment which can be used to satisfy our needs, provided it is technologically accessible, economically feasible and culturally acceptable (1); Examples are water, soil, forests and minerals (1).	1+1
13	Sustainable development means development that meets present needs without damaging the environment (1); It does not compromise the ability of future generations to meet their own needs (1).	1+1
14	Bhangar is older alluvium, deposited away from the river floodplain (1); It has a higher concentration of kankar nodules and is less fertile (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Shelter belts are rows of trees planted to check wind speed and protect the soil (1); They help in stabilising sand dunes and protecting agriculture, e.g., in western India (1); They have helped in stabilising the desert in western India (1).	1+1+1
16	Black soil is made of volcanic lava and is clayey, found in the Deccan Trap region (1); It retains moisture, becomes sticky when wet and develops cracks when dry (1); It is rich in lime, iron, magnesia and potash, and ideal for cotton (1).	1+1+1
17	Gandhiji said there is enough for everybody's need but not for anybody's greed (1); He blamed the greedy and the exploitative nature of modern technology for resource depletion (1); He favoured production by the masses instead of mass production (1).	1+1+1
	OR: Sustainable development means progress without harming the environment, so that future generations can also meet their needs (1); It was highlighted by the Brundtland Commission report and the 1992 Rio Summit (1); It requires careful planning and conservation of resources (1).	1+1+1
18	Overgrazing, deforestation and mining are major causes of land degradation (1); Over-irrigation leads to waterlogging and salinity (1); Industrial effluents and mineral processing dust also damage land (1).	1+1+1
	OR: The first stage is identification and inventory of resources by surveying, mapping and estimation (1); The second stage is evolving a planning structure with appropriate technology, skill and institutions (1); The third is matching resource development plans with national development plans (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

19	Land is used for forests, agriculture, pastures, non-agricultural uses and barren wasteland (1); India's total geographical area is about 3.28 million sq km (1); Land use depends on physical factors like topography, soil and climate (1); Human factors such as population density, technology and culture also matter (1); Net sown area is the land actually cropped (1).	1+1+1+1+1
	OR: Alluvial soil is deposited by rivers in the northern plains and is very fertile (1); It is rich in potash, phosphoric acid and lime, and suits sugarcane, rice and wheat (1); Black soil is formed from lava in the Deccan and is made of fine clay (1); It retains moisture, is sticky when wet and cracks when dry (1); It is ideal for cotton, so called black cotton soil (1).	1+1+1+1+1
20	Sustainable development means meeting present needs without harming future generations (1); The Brundtland Commission popularised the idea (1); The 1992 Rio de Janeiro Earth Summit adopted Agenda 21 to tackle environmental damage and poverty (1); Over 100 heads of state attended (1); Gandhi's warning against greed expresses the same concern (1).	1+1+1+1+1
	OR: Individual resources are owned privately, such as plots and houses (1); Community resources are common to all members, such as village pastures and public parks (1); National resources are within a country's political boundaries and the government can acquire them (1); International resources are oceanic resources beyond 200 nautical miles regulated by international institutions (1); Each type needs responsible management (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing up classifications: **renewable/non-renewable** (by exhaustibility) and **biotic/abiotic** (by origin) are different; coal is both biotic and non-renewable.
- Confusing **stock** (not yet usable for lack of technology) with **reserves** (the part of stock that can be put to use with existing technology).
- Swapping **bangar** (older alluvium, kankar nodules) and **khadar** (newer alluvium, more fertile).
- Writing that red soil is formed from alluvium; it develops on **crystalline igneous and metamorphic rocks**, and its colour is due to iron compounds.
- Calling black soil 'alluvial' or forgetting it is poor in phosphoric content though rich in lime, potash and magnesium.
- Dating the Earth Summit and **Agenda 21** at Rio de Janeiro wrongly; both are **1992**, not Kyoto or Paris.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER

ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Resources and Development (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** other than current fallow	Land use
2	**(a)** the Rio Earth Summit	Sustainable goals
3	**(c)** surveying and mapping	First stage
4	**(c)** sheet erosion	Surface wash
5	**(c)** the area sown, counting land sown more than once only once	Actual cultivation
6	**(d)** national resources	State owns
7	**(a)** alluvial soil	About 40% area
8	**(a)** crystalline igneous rocks	Colour due to iron
9	**(d)** regur soil	Cotton soil
10	**(c)** cotton	Deccan trap area

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Strip cropping is growing crops in alternate strips with strips of grass left in between (1); It breaks the force of wind and water and reduces soil erosion (1).	1+1
12	Leaching is the washing away of nutrient-rich top layers of soil by heavy rain (1); It leaves the soil poor in nutrients and humus, as in laterite soils (1).	1+1
13	Gully erosion occurs when running water cuts deep channels in clayey soils (1); The land becomes unfit for cultivation and is called bad land or ravines, e.g. the Chambal (1).	1+1
14	Khadar is newer alluvium, deposited by rivers near their floodplains (1); It has finer particles, is more fertile and gets replenished every year (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Red soil develops on crystalline igneous rocks in areas of low rainfall in the Deccan plateau (1); It gets its colour from the diffusion of iron in crystalline rocks and looks yellow when hydrated (1); It is found in Odisha, Chhattisgarh and parts of the Western Ghats foothills (1).	1+1+1
16	Forest soils are found in hilly and mountainous regions with sufficient rain forests (1); They are loamy and silty in valley sides and coarse-grained in the upper slopes (1); In snow-covered areas they are acidic with low humus content (1).	1+1+1
17	Alluvial soil is the most widespread, deposited by the Indus, Ganga and Brahmaputra rivers (1); It is rich in potash, phosphoric acid and lime and is highly fertile (1); It is ideal for sugarcane, paddy, wheat and other cereals and pulses (1).	1+1+1
	OR: Contour ploughing means ploughing along contour lines on a slope, which decelerates the flow of water down the slope (1); Terrace farming cuts steps on slopes to check run-off, practised in the Western and Central Himalayas (1); Both reduce soil erosion on slopes (1).	1+1+1
18	On the basis of ownership, resources are individual, community, national and international (1); Individual resources include plots, houses and gardens, while community resources include village commons, burial grounds and public parks (1); National resources include all within the political boundaries; international resources are oceanic resources beyond 200 nautical miles regulated by international institutions (1).	1+1+1
	OR: Land uses include forests, land not available for cultivation, and other uncultivated land excluding fallow (1); Fallow lands (current fallow and other than current fallow) are left uncultivated for some time (1); Net sown area is the area under crops (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

19	Soil erosion is the removal of soil by wind and water faster than it forms (1); It takes forms such as gully erosion and sheet erosion (1); Contour ploughing slows water flow on slopes (1); Terrace farming and strip cropping reduce run-off (1); Shelter belts check wind erosion (1).	1+1+1+1+1
	OR: A thing becomes a resource only when it is technologically accessible, economically feasible and culturally acceptable (1); For example, water became a resource when technology allowed hydroelectricity (1); Resources are not, they become (1); Culture decides use, e.g. different groups use materials differently (1); Human effort is thus an essential part of resource creation (1).	1+1+1+1+1
20	Resources are unevenly distributed across regions, so planning is essential (1); Some regions are rich in minerals but poor in water and infrastructure (1); Stage 1 is identification and inventory by surveying and mapping (1); Stage 2 is evolving a planning structure with technology, skill and institutions (1); Stage 3 is matching resource plans with national development plans (1).	1+1+1+1+1
	OR: Red soil develops on crystalline rocks in low-rainfall areas and gets its colour from iron (1); Yellow soil forms when red soil is hydrated (1); Laterite soil forms in hot, humid areas with heavy rain and intense leaching (1); It supports tea, coffee and cashew with fertilisers (1); Arid soils are sandy and saline, found in Rajasthan, with low humus, and need irrigation (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing up classifications: **renewable/non-renewable** (by exhaustibility) and **biotic/abiotic** (by origin) are different; coal is both biotic and non-renewable.
- Confusing **stock** (not yet usable for lack of technology) with **reserves** (the part of stock that can be put to use with existing technology).
- Swapping **bangar** (older alluvium, kankar nodules) and **khadar** (newer alluvium, more fertile).
- Writing that red soil is formed from alluvium; it develops on **crystalline igneous and metamorphic rocks**, and its colour is due to iron compounds.
- Calling black soil 'alluvial' or forgetting it is poor in phosphoric content though rich in lime, potash and magnesium.
- Dating the Earth Summit and **Agenda 21** at Rio de Janeiro wrongly; both are **1992**, not Kyoto or Paris.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER

ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Resources and Development (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** biotic and abiotic	Origin
2	**(a)** gullies	Gully erosion
3	**(a)** 1992	UN conference
4	**(d)** barren land and land put to non-agricultural uses	Land use category
5	**(a)** non-renewable	Minerals, fossil fuels
6	**(c)** 16%	Pressure on land
7	**(d)** reserves	In reserve
8	**(d)** potential resources	Wind in Rajasthan
9	**(b)** developed resources	In use
10	**(d)** contour ploughing	Conservation

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Potential resources are resources found in a region but not yet utilised (1); For example, wind and solar energy in Rajasthan and Gujarat, which have not been developed fully (1).	1+1
12	Net sown area is the physical extent of land on which crops are sown and harvested (1); Land sown more than once is counted only once (1).	1+1
13	Black soil is also called regur soil and is ideal for growing cotton (1); It is formed by weathering of volcanic Deccan lava rocks and is rich in clay (1).	1+1
14	Agenda 21 was adopted at the Rio de Janeiro Earth Summit in 1992 by world leaders (1); It aims to combat environmental damage, poverty and disease through global cooperation and local plans (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Arid soils are red to brown in colour and sandy in texture, found in Rajasthan (1); They are saline and lack humus and moisture (1); A kankar layer at the bottom restricts water infiltration; irrigation can make them cultivable (1).	1+1+1
16	Stock is the materials that can satisfy human needs but humans lack the appropriate technology to access, e.g., hydrogen in water (1); Reserves are the subset of stock that can be put into use with existing technology but their use has not begun, e.g., river water in dams (1); Developed resources have been surveyed, and their quality and quantity determined for use (1).	1+1+1
17	Wind erosion removes loose soil in deserts, and sheet erosion removes the top layer over large areas (1); Gully erosion cuts deep channels and leaves ravines (1); These make land unfit for cultivation and damage soil fertility (1).	1+1+1
	OR: Resources are unevenly distributed over the country, so regional planning is needed (1); Some resources are limited or non-renewable and wasteful use would exhaust them (1); Planning ensures balanced development and sustainable use (1).	1+1+1
18	On the basis of origin, resources are biotic or abiotic (1); Biotic resources are obtained from the living biosphere, e.g., forests, animals and fisheries (1); Abiotic resources are composed of non-living things, e.g., rocks, metals and water (1).	1+1+1
	OR: Laterite soil develops in tropical and subtropical areas with high temperature and heavy rain (1); Intense leaching leaves it poor in humus and nutrients (1); With manures and fertilisers it supports tea, coffee and cashew (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

19	Land is degraded by mining, overgrazing, over-irrigation and industrial waste (1); Mining affects Jharkhand, Chhattisgarh, Madhya Pradesh and Odisha (1); Overgrazing affects Gujarat, Rajasthan and Maharashtra, and over-irrigation affects Punjab and Haryana (1); Remedies include afforestation, proper grazing, shelter belts and stabilising sand dunes (1); Mining and industrial effluents must be controlled (1).	1+1+1+1+1
	OR: Soils are mostly alluvial, covering the northern plains and coastal deltas (1); Black soil covers the Deccan plateau (1); Red and yellow soils cover the eastern and southern plateaus, and laterite the Western Ghats and hill areas (1); Arid soils occur in Rajasthan, and forest soils in hills (1); Soils support farming, which feeds the population (1).	1+1+1+1+1
20	By origin, resources are biotic (forests, animals, fisheries) and abiotic (rocks, minerals, metals) (1); By exhaustibility, they are renewable (solar and wind energy, water) and non-renewable (minerals, fossil fuels) (1); By ownership, they are individual, community, national and international (1); By status of development, they are potential, developed, stock and reserves (1); Classification helps in planning and conserving them (1).	1+1+1+1+1
	OR: India has 2.4 per cent of the world's area but supports about 16 per cent of the population (1); Pressure on land is very high (1); Land degradation reduces productivity (1); Sustainable use ensures food security for future generations (1); Conservation measures keep soil fertile (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing up classifications: **renewable/non-renewable** (by exhaustibility) and **biotic/abiotic** (by origin) are different; coal is both biotic and non-renewable.
- Confusing **stock** (not yet usable for lack of technology) with **reserves** (the part of stock that can be put to use with existing technology).
- Swapping **bangar** (older alluvium, kankar nodules) and **khadar** (newer alluvium, more fertile).
- Writing that red soil is formed from alluvium; it develops on **crystalline igneous and metamorphic rocks**, and its colour is due to iron compounds.
- Calling black soil 'alluvial' or forgetting it is poor in phosphoric content though rich in lime, potash and magnesium.
- Dating the Earth Summit and **Agenda 21** at Rio de Janeiro wrongly; both are **1992**, not Kyoto or Paris.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER